

LAW2 ENDCAP REDESIGNS

Andrew:

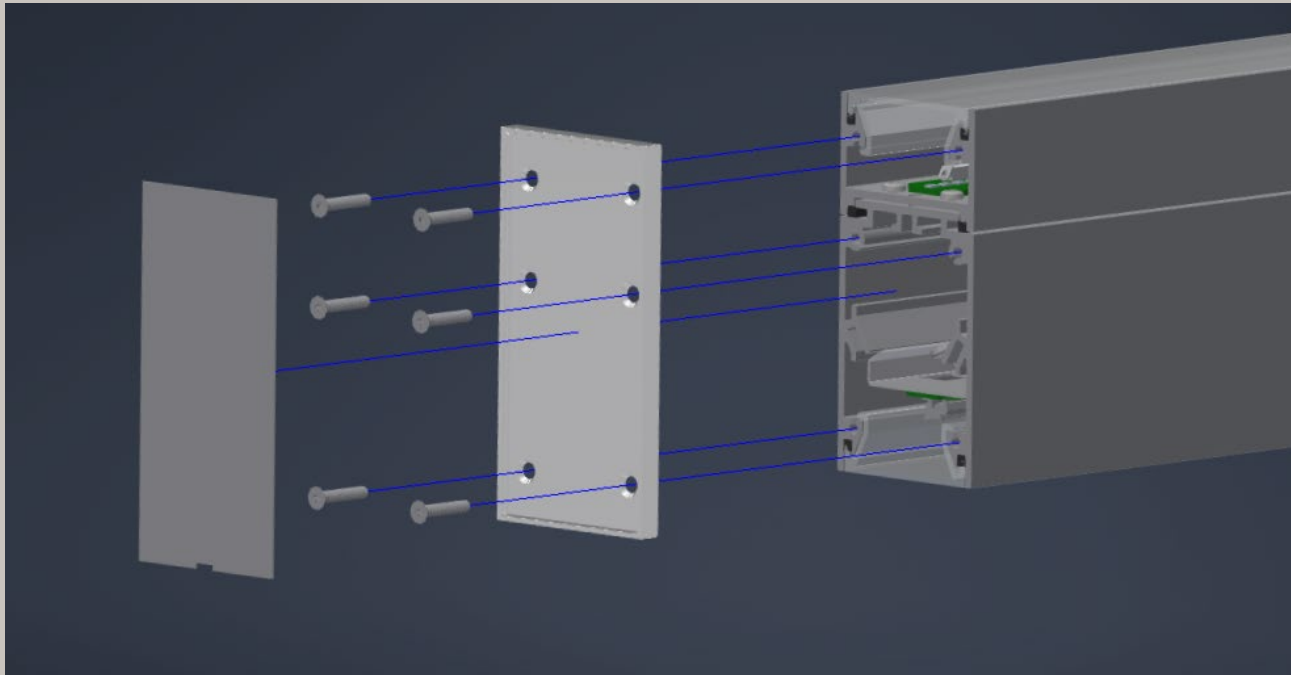
Up Down & Up Down Drop

Lucas:

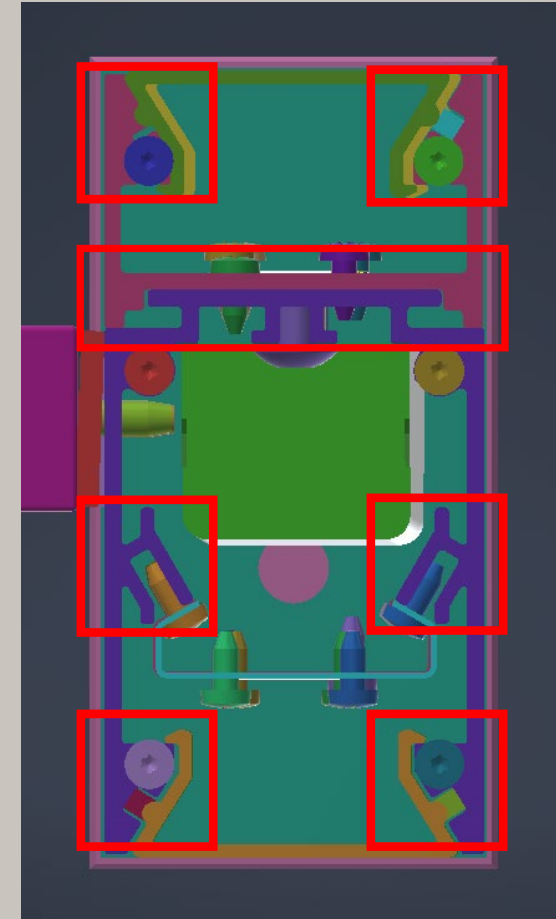
Surf Pend & Surf Pend Drop

OLD END CAPS

- Gasket compression based on torque (Inconsistent)
- Very thin and weak (susceptible to warping)
- Includes separate back plate (\$\$\$\$\$)



Points of contact:

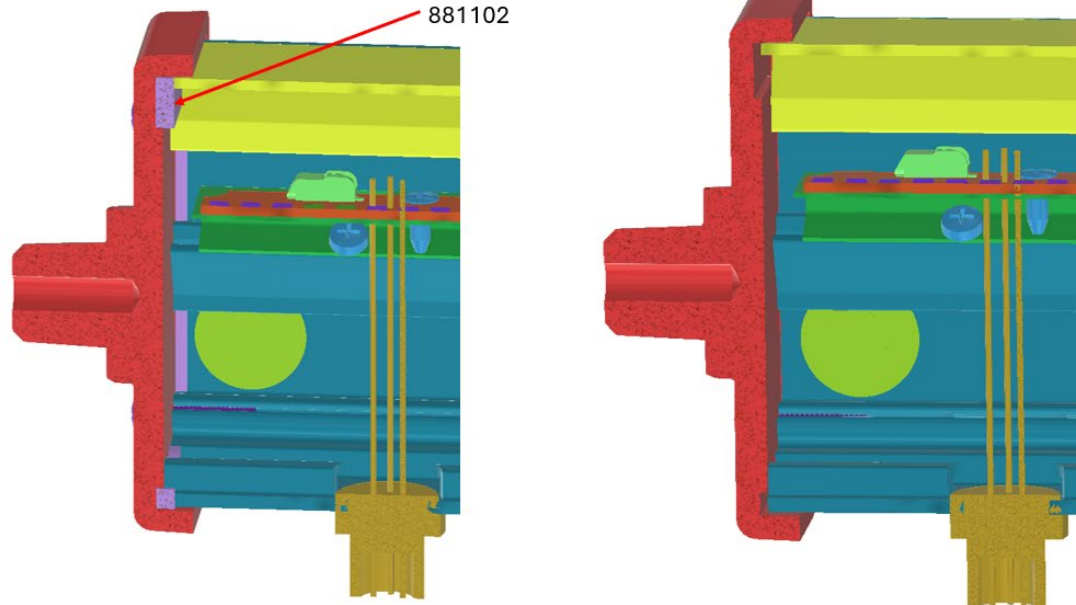


WHY?

- To Match the “ADJUSTABLE ARM END CAP HIRTH”
 - It can provide consistent compression of the gasket due to the mechanical stop
 - Has great durability and strength

889337.IAM View: 6. END CAP

Flush lens, Direct only



ITERATIONS

ITERATIONS – UP DOWN & UP DOWN DROP

- 1st (Before learning about die casting)
 - Pros – Minimal added material, simple
 - Cons – Small amount of contact area, thin walls (tricky to cast and likely to crush)



- 3rd
 - Pros – more contact area at main points, small amount of added material
 - Cons – Flimsy, had little bend support on ends; feared tight screws would bend the cap



ITERATIONS – UP DOWN & UP DOWN DROP

- 4th (learned about cost of aluminum; \$3/lb)
 - Pros – Better bend support, full contact with extrusion
 - Cons – Still very flimsy, little overlap, not much room for gasket around holes, no draft angles



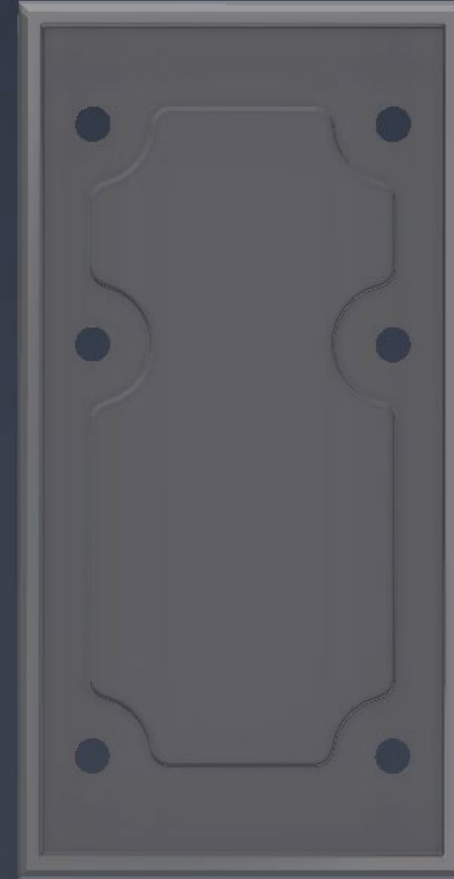
- 9th (FLAT BACK is introduced)
 - Pros – Great bend support, strong and durable, more overlap (0.200”), space for gasket, separate cover is unneeded
 - Cons – heavier/more expensive, FLAT BACKs aren't flat (ironic) due to O-ring and screw



RESULTS

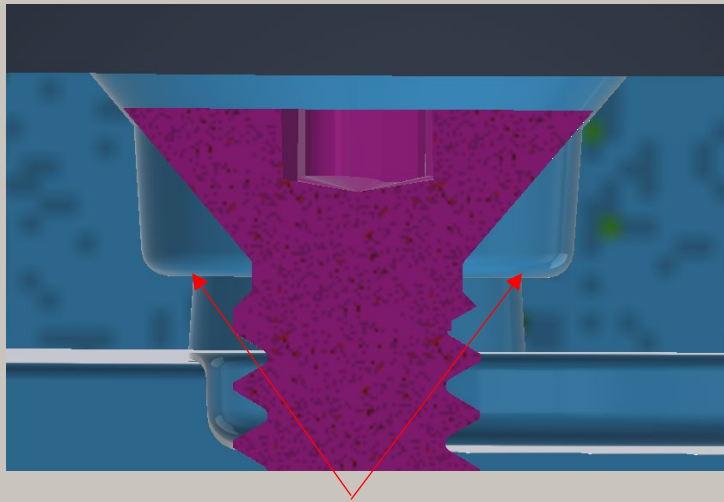
RESULTS – UP DOWN & UP DOWN DROP

- Increased strength
 - Increased wall thickness (0.060" to ≈ 0.110 ")
 - Increased plate thickness (0.055" to 0.075")
- Improved capabilities
 - Taller walls (0.305" to 0.450") to allow for more depth (0.125" to 0.250")
 - Wider holes (0.150" to 0.183")
 - Added a raised surface/mechanical stop (0.050")



RESULTS – FLAT BACKS

- Same dimensions as basic designs besides no depth cut on the back
- Additional back cover is unnecessary (Large cost reduction)
- New hole design to compensate for O-ring (20% compression)

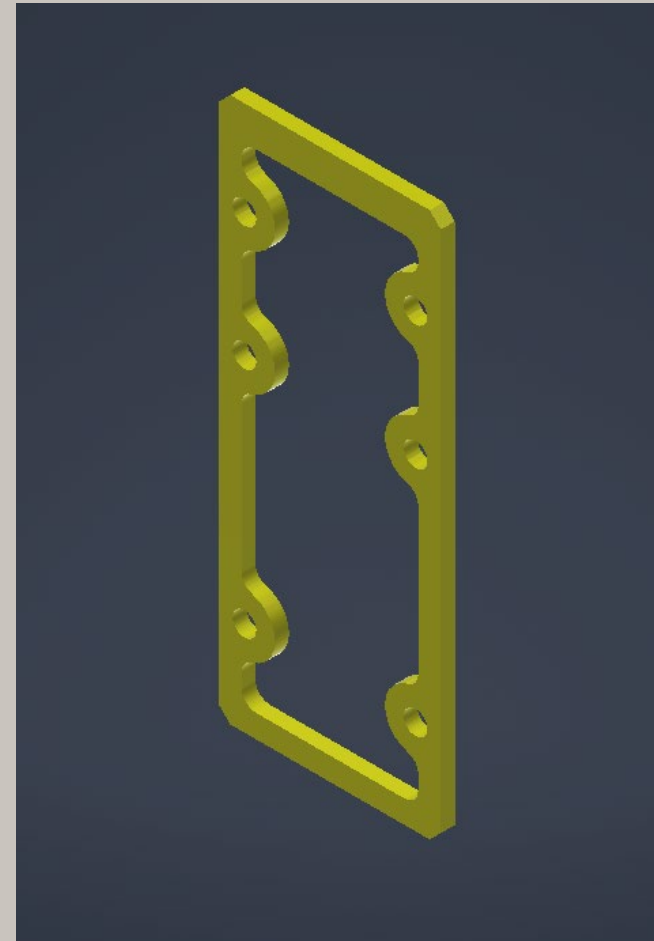
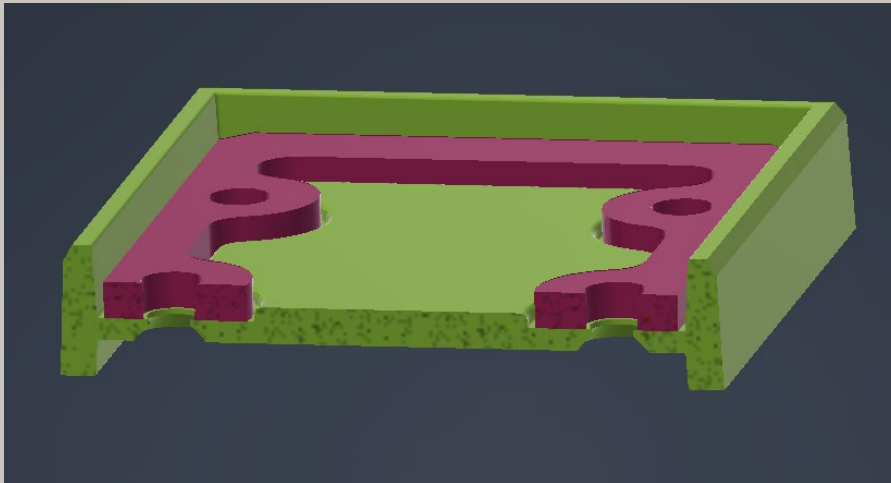


Gap for O-ring



RESULTS – UP DOWN AND UP DOWN DROP GASKET

- Matched thickness from “ADJUSTABLE ARM END CAP HIRTH” (0.125”, 40% compression)
- 0.015” offset from walls and holes
- Chamfered corners for easy implementation



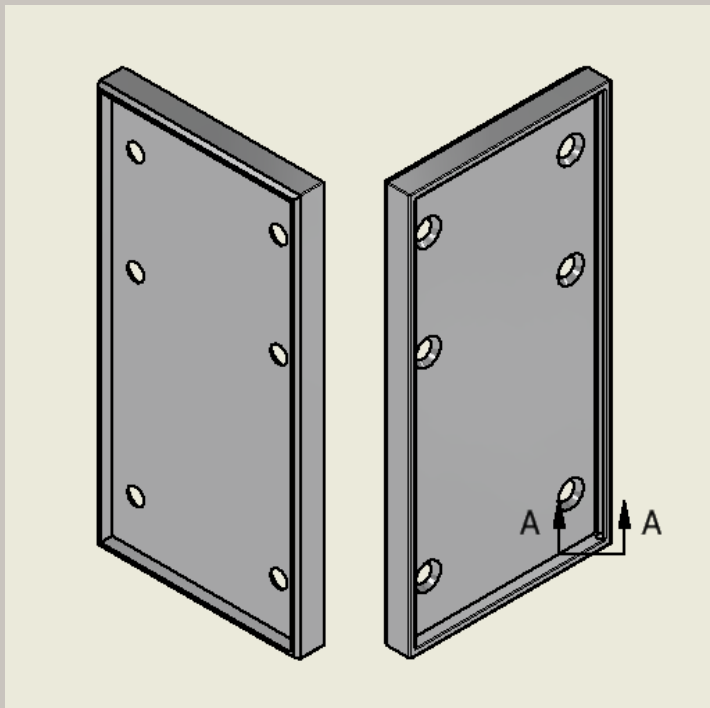
RESULTS – FINAL PROTOTYPES



COST COMPARISONS

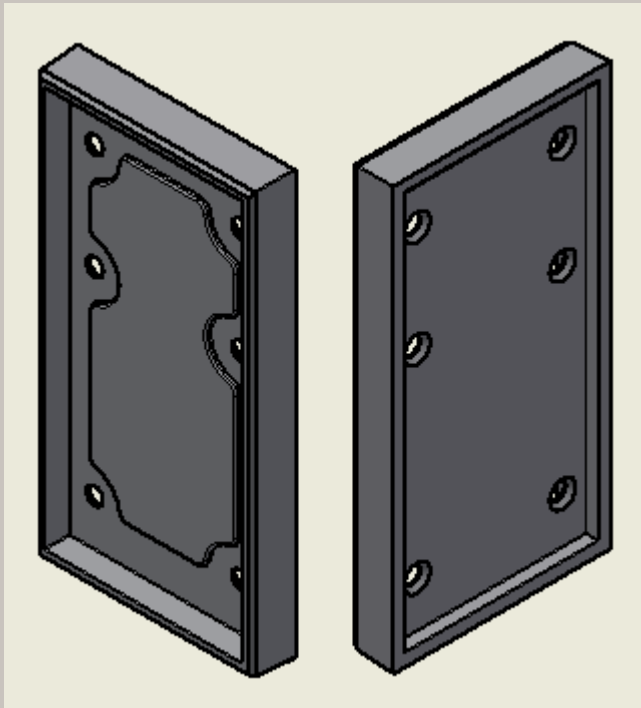
COST COMPARISON – UP DOWN

Old (803727)



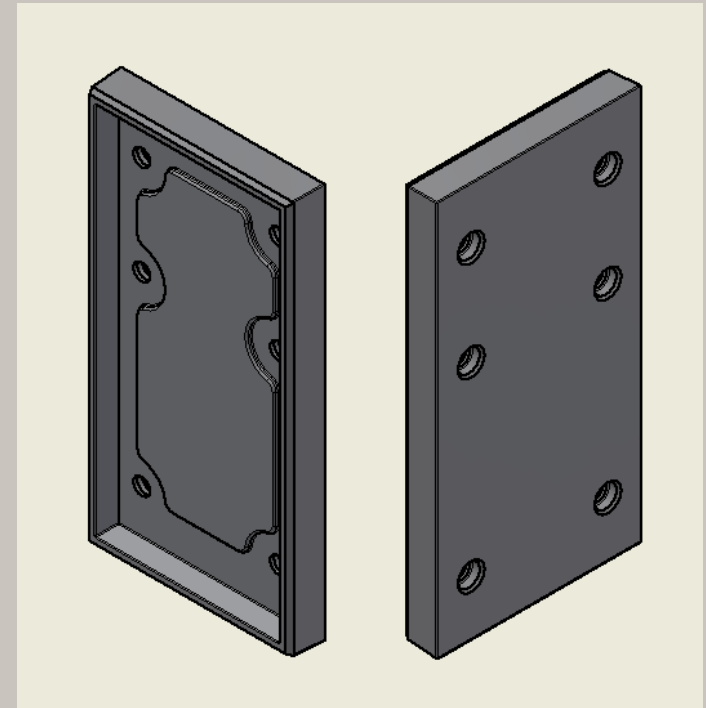
Mass = 0.064 (≈\$0.20)

New (918025)



Mass = 0.145 (≈\$0.44; +\$0.22)

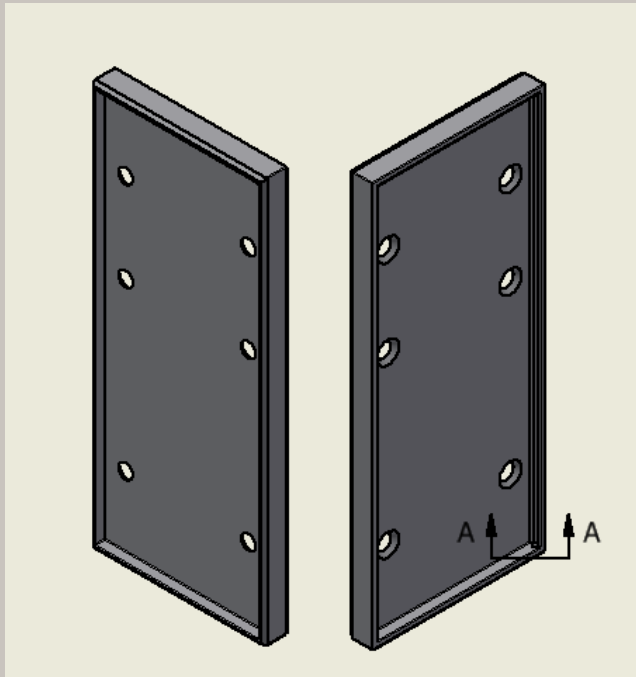
New FLAT BACK (918466)



Mass = 0.195 (≈\$0.59; - \$0.99)

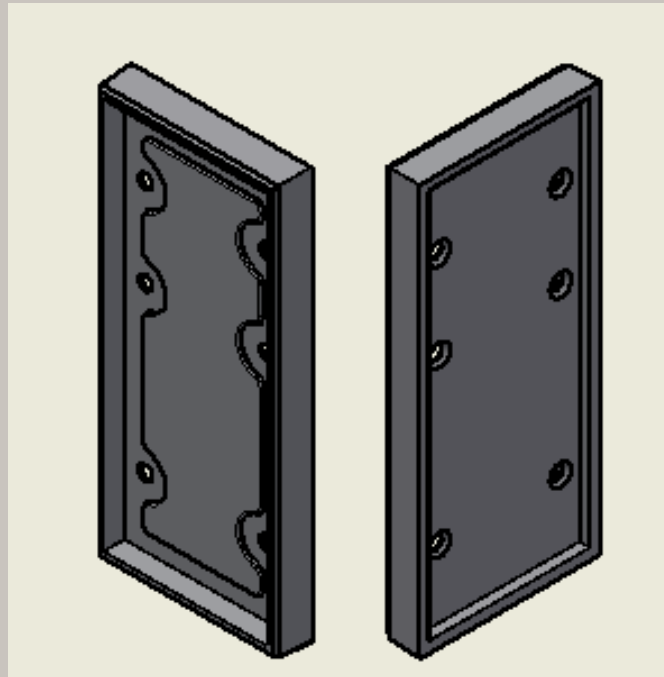
COST COMPARISON – UP DOWN DROP

Old (803725)



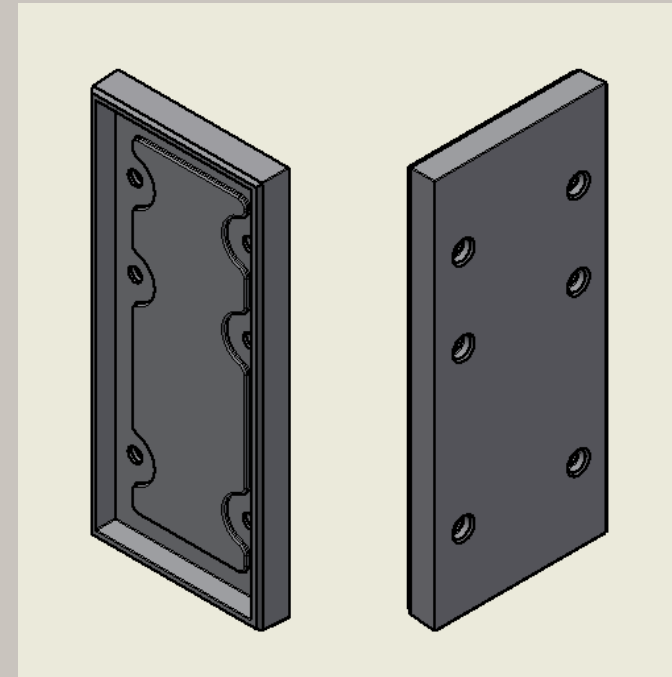
Mass = 0.075 ($\approx$ \$0.23)

New (918016)



Mass = 0.168 ($\approx$ \$0.50; +\$0.27)

New FLAT BACK (918470)



Mass = 0.230 ($\approx$ \$0.69; -\$1.34)

WHAT WAS LEARNED

LEARNED ABOUT:

- Gasket compression
- Die casting
 - How it works
 - Common practice/processes
 - Draft Angles
- Design process and thinking ahead
- Drawing details and standards
 - Tolerances
 - Stack-ups
- Metal costs
- 3D modeling application uses
 - Inventor
 - Vault
 - JDE
- Importance of prototyping
- Communication